SOFTWARE REQUIREMENTS SPECIFICATION

CampusCart

A Campus-Exclusive Online Marketplace with Direct Messaging

22MCCE34

22MCCE37

1. INTRODUCTION

1.1 Purpose

This Software Requirements Specification (SRS) document describes the functional and non-functional requirements for CampusCart, a campus-exclusive online marketplace system. The purpose of this document is to provide a comprehensive and unambiguous description of the system to be developed, serving as a reference for developers, testers, project managers, and stakeholders throughout the software development lifecycle.

This SRS covers all aspects of the system including user interfaces, functional capabilities, performance constraints, and system attributes required to build a secure, reliable, and scalable campus marketplace platform.

1.2 Scope

CampusCart is a centralized digital marketplace designed exclusively for college students to buy and sell items within their campus community. The system replaces fragmented communication channels (such as WhatsApp groups) and protects student privacy by restricting access to verified college email IDs only.

The system includes the following major functions:

- User registration and authentication using college-verified email IDs
- Product listing, browsing, and searching across predefined categories
- Direct in-app messaging between buyers and sellers for product inquiries and negotiations
- Auction module for time-based competitive bidding
- Listing management for sellers
- User and platform administration by system administrators

CampusCart does not cover general public e-commerce, delivery logistics, physical payment handling, or integration with external retail or payment platforms.

1.3 Definitions, Acronyms, Abbreviations

Term / Acronym	Definition
SRS	Software Requirements Specification
UC	Use Case
UI	User Interface
DB	Database
API	Application Programming Interface
DFD	Data Flow Diagram
Admin	System Administrator

OTP	One-Time Password
HTTPS	HyperText Transfer Protocol Secure
SMTP	Simple Mail Transfer Protocol
Buyer	A registered student who searches for products and contacts sellers
Seller	A registered student who posts products for sale or auction and manages buyer inquiries
Administrator	Platform manager with access to user and content management
Listing	A product posted by a Seller on the platform
Auction	A time-limited bidding process for a listed product
MessageThread	A private communication channel between a Buyer and a Seller regarding a specific listing

1.4 References

- Use Case Specifications Document – Online Marketplace
- DFD Level 0 and Level 1 Diagrams – CampusCart (Revised – Messaging Architecture)
- Use Case Diagram – Online Marketplace System (Revised – Messaging Architecture)
- IEEE Std 830-1998: IEEE Recommended Practice for Software Requirements Specifications

1.5 Overall Description

CampusCart addresses a common problem faced by college students who rely on informal, privacy-compromising channels to buy and sell second-hand items. The platform provides a structured, secure, and campus-restricted digital marketplace that centralizes listings, hides personal contact details, and enables direct in-app messaging between buyers and sellers similar to the OLX model eliminating the need to expose personal phone numbers or emails.

The system operates as a web-based application accessible via browsers and mobile devices. It integrates three primary user roles Buyer, Seller, and Administrator each with distinct access rights and functional capabilities. The platform relies on college email verification to restrict membership, ensuring a trusted community environment.

2. OVERALL DESCRIPTION

2.1 Product Perspective

CampusCart is a standalone web application that functions as a campus-exclusive digital marketplace. It is not a subsystem of any larger existing platform. The system interacts with the following external entities:

- Students (Buyers and Sellers): Interact through a web/mobile browser interface
- College Email Server: Used for registration verification via SMTP-based email validation
- Cloud Storage Service: For storing product images and user-uploaded files

The system's architecture is divided into three internal modules as depicted in the DFD Level 1 diagram: Messaging Module, Product Module, and User Management Module. These modules interact with each other to deliver a cohesive user experience.

2.2 Product Functions

The major functions of CampusCart are summarized below:

- User Registration and Authentication: College email-based sign-up with email verification. Secure login with session management and account lockout after multiple failed attempts.
- Product Listing Management: Sellers can post products with titles, descriptions, prices, categories, and images. Listings can be edited, deactivated, or deleted by the seller.
- Product Search and Discovery: Buyers can search products using keywords or browse by category. Results are sorted by relevance.
- Direct Messaging: Buyers can initiate a private chat with sellers directly from the product page. Both parties can view and manage their active message threads. All communication remains within the platform, protecting personal contact details.
- Auction Module: Sellers can create time-limited auctions. Buyers can place bids; the highest bidder at auction end wins.
- Notifications: Email alerts for key events including registration, listing confirmation, new messages, and auction results.
- Admin Panel: Administrators can manage user accounts (view, block, delete), moderate listings, and access user reports.

2.3 User Characteristics

CampusCart serves three distinct user types:

Buyer

College students who wish to inquire about and purchase second-hand items. Buyers are assumed to have basic internet literacy and access to a web browser or mobile device. They may not have prior experience with structured marketplace platforms but are familiar with general messaging and browsing concepts.

Seller

College students who wish to sell items. Sellers must be comfortable uploading product images, writing product descriptions, and responding to buyer inquiries via in-app messaging. They may play both the Buyer and Seller role simultaneously on the platform.

Administrator

A designated system administrator responsible for managing the platform. Administrators are assumed to have moderate technical proficiency, ability to navigate admin dashboards, and understanding of platform policies. Only administrators have access to user management and reporting features.

2.4 User Constraints

- Access is restricted to users with a valid, verified college email domain.
- All users must be enrolled students or authorized staff of the associated college.
- Product categories are predefined; users cannot create new categories.
- Restricted or prohibited items (as per platform policy) may not be listed.
- Auction bids are binding; the winning bidder is obligated to complete the purchase.
- Personal contact information (phone numbers, external email) must not be shared in product listings or messages.

2.5 Assumptions and Dependencies

Assumptions:

- Students have access to a stable internet connection and a web browser or mobile device.
- The college provides a consistent email domain for all enrolled students.
- The email verification service (SMTP server) is available and operational.
- Cloud storage services are accessible for image hosting.

Dependencies:

- Dependency on college email infrastructure for user verification.
- Dependency on cloud storage services for product image storage.
- The system assumes a web server and relational database are provisioned and maintained by the college IT department or a cloud provider.

2.6 Apportioning Requirements

The following requirements are identified for future versions of CampusCart

- Mobile native application (iOS/Android) – Phase 2
- Integrated payment gateway and checkout flow – Phase 2
- Seller reputation and rating system – Phase 2
- Integration with campus ID cards for enhanced verification – Phase 3
- Multi-campus support across different institutions – Phase 3

3. SPECIFIC REQUIREMENTS

3.1 Interface Requirements

3.1.1 External Interface

The system interfaces with the following external entities:

- Buyer Interface: Web-based UI accessible via browser. Allows product search, initiating and viewing message threads with sellers, and messaging inbox management.
- Seller Interface: Web-based UI for posting products, managing listings, and managing buyer inquiries via messaging.
- Administrator Interface: Admin dashboard for user management, content moderation, and report generation.
- Email Notification Interface: System sends automated emails to users at key lifecycle events using SMTP.

3.1.2 Hardware Interface

CampusCart is a web-based system and does not require any specialized hardware. The following hardware is assumed:

- Client Side: Any device (desktop, laptop, tablet, or smartphone) with a modern web browser and internet connectivity.
- Server Side: A web server and database server hosted on-premises or on a cloud platform (e.g., AWS, Azure, or similar). Minimum recommended configuration is 4 vCPUs, 8 GB RAM, and 100 GB scalable storage.
- Storage: Cloud object storage for product images (e.g., AWS S3 or equivalent).

3.1.3 Software Interface

The system interacts with the following software components:

- Web Browser: Compatible with modern browsers (Chrome, Firefox, Safari, Edge).
- Database Management System: Relational database (e.g., MySQL or PostgreSQL) for storing user, product, message, and auction data.
- Email Service: SMTP-compatible email service (e.g., SendGrid, AWS SES) for sending verification and notification emails.
- Cloud Storage: Object storage API for uploading and serving product images.
- Web Framework: Server-side web framework (e.g., Django, Node.js/Express) for application logic.
- WebSocket Library: A server-side WebSocket implementation (e.g., Socket.IO or Django Channels) for real-time message delivery.

3.1.4 Communication Interface

- All communication between the client and server must use HTTPS (TLS 1.2 or higher) to ensure data encryption in transit.
- Email notifications are transmitted using the SMTP protocol via the integrated email service provider.
- Real-time messaging data is transmitted using WebSockets or RESTful polling over HTTPS.
- API responses are formatted in JSON.

3.2 Functional Requirements

3.2.1 Use Case Model / Information Flows

The system is modeled using Use Case Diagrams and Data Flow Diagrams (DFDs). The following summarizes the key information flows:

DFD Level 0 (Context Diagram):

Three external entities interact with the Marketplace System: Buyer, Seller, and Administrator. The Buyer sends product search requests and receives search results and product details. The Buyer also initiates and receives messages. The Seller sends product listings and updates, receives listing confirmations, and exchanges messages with buyers. The Administrator sends user management instructions and receives reports and user data.

DFD Level 1:

The Marketplace System decomposes into three internal modules: Messaging Module, Product Module, and User Management Module. The Buyer interacts with the Messaging Module (initiate chat, view inbox) and the Product Module (search, view items). The Seller interacts with the Product Module (add products, manage listings) and the Messaging Module (respond to buyer inquiries). The Administrator interacts with the User Management Module (manage users).

Use Case Diagram:

The system identifies three actors: Buyer, Seller, and Administrator. The Buyer can perform: Search for Product, View Product Details, Add Product to Cart, and Message Seller. The Seller can perform: Post Product for Sale and Manage Product Listings. The Administrator can perform: Manage Users.

3.2.2 Use Case Specifications / Process Descriptions

UC01 – Register Account

Use Case ID	UC01
Name	Register Account
Actors	Buyer, Seller
Description	Allows a user to register using their college email address and a password.

Preconditions	User must possess a valid college email address. User must not be previously registered.
Post Conditions	A new inactive account is created. A verification email is sent to the user.
Normal Flow	1. User navigates to the registration page.2. User enters name, college email, and password.3. System validates the email domain against the approved college domain.4. System stores the new user record as inactive.5. System sends a verification email with an activation link.6. User clicks the link; account is activated.
Alternate Flows	A1. Email already registered: System displays an error message and prompts login.A2. Invalid email domain: System rejects registration and informs the user.
Exceptions	E1. Email service failure: System displays an error; account is not created.
Priority	High
Business Rules	Only college domain email addresses are permitted for registration.

UC02 – Login

Use Case ID	UC02
Name	Login
Actors	Buyer, Seller, Administrator
Description	Allows a registered and verified user to log into the system.
Preconditions	Account must be registered and email-verified.
Post Conditions	A user session is created; the user is redirected to their dashboard.
Normal Flow	1. User enters email and password.2. System validates credentials against the database.3. System creates a session token and redirects user to the dashboard.
Alternate Flows	A1. Wrong password: System displays an error. After three failed attempts, the account is locked.
Exceptions	E1. Server unavailable: System displays a service unavailable error.
Priority	High
Business Rules	Three consecutive failed login attempts result in a temporary account lockout.

UC03 – Search Product

Use Case ID	UC03
Name	Search Product
Actors	Buyer
Description	Allows a Buyer to search for products using keywords or by selecting a category.

Preconditions	None (search is available to all visitors; full details require login).
Post Conditions	A list of matching products is displayed to the user.
Normal Flow	1. Buyer enters a keyword or selects a category.2. System queries the product database.3. System displays matching results sorted by relevance.
Alternate Flows	A1. No results found: System displays a 'No products found' message.
Exceptions	E1. Database error: System displays an error message.
Priority	Medium
Business Rules	Results are sorted by relevance. Inactive listings are excluded from results.

UC04 – Post Product

Use Case ID	UC04
Name	Post Product for Sale
Actors	Seller
Description	Allows a logged-in Seller to create a product listing on the platform.
Preconditions	Seller must be logged in with an active, verified account.
Post Conditions	A new product listing is created and visible to Buyers.
Normal Flow	1. Seller navigates to 'Post Product'.2. Seller enters product title, description, price, category, and uploads images.3. System validates input and checks against restricted product policies.4. System saves the listing and sends a confirmation notification.
Alternate Flows	A1. Image upload fails: System notifies the user; listing can still be saved without images.
Exceptions	E1. Storage server error: Listing cannot be saved; system displays an error.
Priority	High
Business Rules	Restricted or prohibited items are not permitted. Sellers must not include personal contact details in listings.

UC05 – Manage Users

Use Case ID	UC05
Name	Manage Users
Actors	Administrator
Description	Allows the Administrator to review, block, or delete user accounts.
Preconditions	Administrator must be logged in with administrator privileges.
Post Conditions	User account status is updated in the database.

Normal Flow	1. Admin navigates to the User Management panel.2. Admin views the list of registered users.3. Admin selects a user and performs an action (block/unblock/delete).4. System updates the user's status in the database and confirms the change.
Alternate Flows	None
Exceptions	E1. Permission error: Action is rejected; system displays an authorization error.
Priority	Medium
Business Rules	Only users with Administrator privileges can modify user account status.

UC06 – Message Seller

Use Case ID	UC06
Name	Message Seller
Actors	Buyer, Seller
Description	Allows a Buyer to initiate a private message thread with a Seller from a product listing page. Both parties can send and receive messages within the thread.
Preconditions	Buyer must be logged in. The product listing must be active. A message thread between this Buyer and Seller for this product must not already exist (system reuses existing threads).
Post Conditions	A new MessageThread is created (or an existing one is retrieved). The Buyer's message is delivered to the Seller. The Seller receives a notification.
Normal Flow	1. Buyer views a product detail page and clicks 'Message Seller'.2. System checks if a MessageThread already exists between this Buyer, Seller, and product.3. If no thread exists, system creates a new MessageThread record.4. Buyer types and submits a message.5. System saves the message and delivers it to the Seller in real time.6. Seller receives an in-app notification and email alert.7. Seller opens the thread and replies.8. Buyer receives the reply in the messaging inbox.
Alternate Flows	A1. Thread already exists: System opens the existing thread instead of creating a new one.A2. Seller has blocked the Buyer: System informs the Buyer that messaging is unavailable.
Exceptions	E1. Messaging service error: Message is not sent; system displays an error and prompts retry.
Priority	High
Frequency of Use	Very High
Business Rules	Users must not share personal contact information (phone numbers, external emails) within messages. All messages are stored and may be reviewed by Administrators for policy compliance.

3.2.3 Analysis Classes / Data Dictionary

The following data dictionary describes the key data entities in CampusCart:

Entity / Attribute	Data Type	Constraints	Description
User			
user_id	Integer	PK, Auto	Unique user identifier
name	String(100)	Not Null	Full name of the user
email	String(150)	Unique, Not Null	College email address
password_hash	String(256)	Not Null	Encrypted password
role	Enum	Buyer/Seller/Admin	Role assigned to the user
is_verified	Boolean	Default: False	Email verification status
is_active	Boolean	Default: True	Account active/blocked status
created_at	DateTime	Not Null	Account creation timestamp
Product			
product_id	Integer	PK, Auto	Unique product identifier
seller_id	Integer	FK → User	Reference to the seller
title	String(200)	Not Null	Product title
description	Text	Not Null	Product description
price	Decimal(10,2)	Not Null, > 0	Listed price
category	String(100)	Not Null	Product category
image_urls	JSON	Nullable	Array of image storage URLs
is_available	Boolean	Default: True	Availability status
created_at	DateTime	Not Null	Listing creation timestamp
MessageThread			
thread_id	Integer	PK, Auto	Unique thread identifier
product_id	Integer	FK → Product	Product the thread is about
buyer_id	Integer	FK → User	The buyer in the conversation
seller_id	Integer	FK → User	The seller in the conversation
created_at	DateTime	Not Null	Thread creation timestamp
Message			
message_id	Integer	PK, Auto	Unique message identifier
thread_id	Integer	FK → MessageThread	Thread this message belongs to
sender_id	Integer	FK → User	User who sent the message
content	Text	Not Null	Message body text

created_at	DateTime	Not Null	Message send timestamp
Auction			
auction_id	Integer	PK, Auto	Unique auction identifier
product_id	Integer	FK → Product	Product being auctioned
start_price	Decimal(10,2)	Not Null, > 0	Starting bid price
end_time	DateTime	Not Null	Auction end timestamp
winner_id	Integer	FK → User, Nullable	Winning bidder user ID
winning_bid	Decimal(10,2)	Nullable	Final winning bid amount

3.3 Performance Requirements

- The system must support a minimum of 500 concurrent users without significant degradation in response time.
- Product search queries must return results within 2 seconds under normal load conditions.
- Page load time for product listing and detail pages must not exceed 3 seconds on a standard broadband connection.
- In-app messages must be delivered and displayed within 2 seconds under normal load conditions.
- The auction module must accurately reflect bid updates with a maximum delay of 5 seconds.
- The system must achieve at least 99.5% uptime during the academic year.
- Database queries must be optimized with appropriate indexing to handle growth up to 10,000 product listings without performance degradation.

3.4 Logical Database Requirements

The database must satisfy the following requirements:

- A relational database management system (RDBMS) such as MySQL or PostgreSQL must be used.
- All user passwords must be stored as hashed values using a secure algorithm (e.g., bcrypt).
- Data integrity must be enforced using primary keys, foreign keys, and database constraints.
- Database transactions must be used for message thread creation to ensure atomicity and prevent duplicate threads.
- Product images are stored as URLs in the database, with the actual files hosted on a cloud storage service.
- The database must support soft-deletion of user accounts (marking as inactive rather than permanent deletion) to maintain audit trails.
- Indexes must be created on frequently queried columns including email, product category, seller_id, buyer_id, and thread_id.
- The database must be backed up daily with transaction log backups for point-in-time recovery.

3.5 Design Constraints

- The system must be accessible via a web browser without requiring any client-side software installation.
- User registration is strictly limited to verified college email domains; no external email providers are permitted.
- The system must not expose personal contact information (phone numbers, external emails) of any user publicly or through messages.
- Product categories are predefined and managed by the Administrator; users cannot create custom categories.
- The auction timer must be server-side controlled to prevent client-side manipulation.
- The platform must comply with applicable data protection regulations regarding student personal data.
- All third-party integrations (email service, cloud storage) must be replaceable with minimal code changes (loose coupling).
- The system must be developed using standard full-stack web technologies to ensure maintainability by the college IT team.

3.6 Software System Attributes

3.6.1 Reliability

The system must demonstrate consistent and predictable behavior under normal and moderate load conditions. Key reliability requirements include:

- Message thread creation must be atomic; if any part of the thread creation process fails, a full rollback must occur to prevent duplicate or orphaned records.
- The auction module must ensure that bid timestamps are accurately recorded and the correct winner is determined at auction end.
- The system must implement automated health checks and restart services automatically in case of process failure.
- Data loss must be prevented through regular database backups and transaction logging.

3.6.2 Availability

- The system must target a minimum availability of 99.5% during the academic year.
- Scheduled maintenance windows must be communicated to users at least 24 hours in advance.
- The system must display informative error pages during downtime rather than raw error messages.
- Critical services such as login and product search must be prioritized for availability during partial system failures.

3.6.3 Security

- All data in transit must be encrypted using HTTPS (TLS 1.2 or higher).
- User passwords must be hashed using a strong algorithm (e.g., bcrypt with appropriate salt rounds) before storage.

- Access control must be enforced at the server level; API endpoints must validate user roles and session tokens.
- SQL injection, Cross-Site Scripting (XSS), and Cross-Site Request Forgery (CSRF) protections must be implemented.
- User accounts must be locked after three consecutive failed login attempts.
- Student personal contact details must never be exposed through the public API or UI.
- Direct messages must be stored securely and only accessible by the participating buyer, seller, and platform administrators.
- Admin endpoints must be protected with additional authentication layers.

3.6.4 Maintainability

- The codebase must follow modular design principles with clear separation between the Messaging, Product, and User Management modules.
- The system must include comprehensive inline code documentation and API documentation.
- Database migrations must be scripted and version-controlled to facilitate schema updates.
- Third-party integrations (email service, cloud storage) must be abstracted behind interfaces to facilitate replacement without major refactoring.
- Logs must be generated for all significant system events (user actions, errors, message thread creations, and auction events) to support debugging and auditing.

3.6.5 Portability

- The application must be deployable on any major cloud platform (AWS, Azure, GCP) or an on-premises Linux server.
- The front-end must be compatible with modern versions of Chrome, Firefox, Safari, and Edge browsers.
- The system must be responsive and function correctly on mobile browsers without a native app.
- Database independence must be maintained where possible by using an ORM layer that supports multiple RDBMS backends.

4. SUPPORTING INFORMATION

4.1 Appendix A – System Diagrams

The following diagrams support this SRS. Note: Diagram files must be redrawn to reflect the revised messaging-based architecture, removing cart/payment/order flows and incorporating messaging flows.

- Use Case Diagram : Illustrates interactions between Buyer, Seller, and Administrator. Buyer use cases include Search for Product, View Product Details, Add Product to Cart, and Message Seller. Seller use cases include Post Product for Sale and Manage Product Listings. Administrator use case is Manage Users.
- DFD Level 0 Context Diagram : Shows the system as a single process with three external entities. Administrator flows show User Management only (no payment management).
- DFD Level 1: Decomposes the Marketplace System into three subsystems: Messaging Module, Product Module, and User Management Module. Illustrates message flows between Buyer, Seller, and the Messaging Module.

4.2 Appendix B – Use Case Summary Table

UC ID	Use Case Name	Actor(s)	Priority	Frequency
UC01	Register Account	Buyer, Seller	High	High
UC02	Login	Buyer, Seller, Admin	High	Very High
UC03	Search Product	Buyer	Medium	High
UC04	Post Product for Sale	Seller	High	Medium
UC05	Manage Users	Administrator	Medium	Occasional
UC06	Message Seller	Buyer, Seller	High	Very High